K-Nearest Neighbors

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Outline

- Parametric vs Non-Parametric
- k-Nearest Neighbors (kNN): Classification
- kNN: Regression

Parametric vs Non-Parametric Learning

- Sejauh ini model dan algoritma yang kita bahas menggunakan pendekatan parametric:
$$\{h_w \mid w \in \mathbb{R}^d\}$$
- Tidak semua model menggunakan pendekatan seperti itu
- Non-Parametric model:
  Keluarga model lainnya yang memiliki kapasitas yang tinggi pula untuk mengestimasi berbagai fungsi yang hampir tanpa batas
- Contoh: k-Nearest Neighbors (kNN)

k-Nearest Neighbors (kNN): Classification

- Classification: Diberikan input data x:
- Langkah-1:

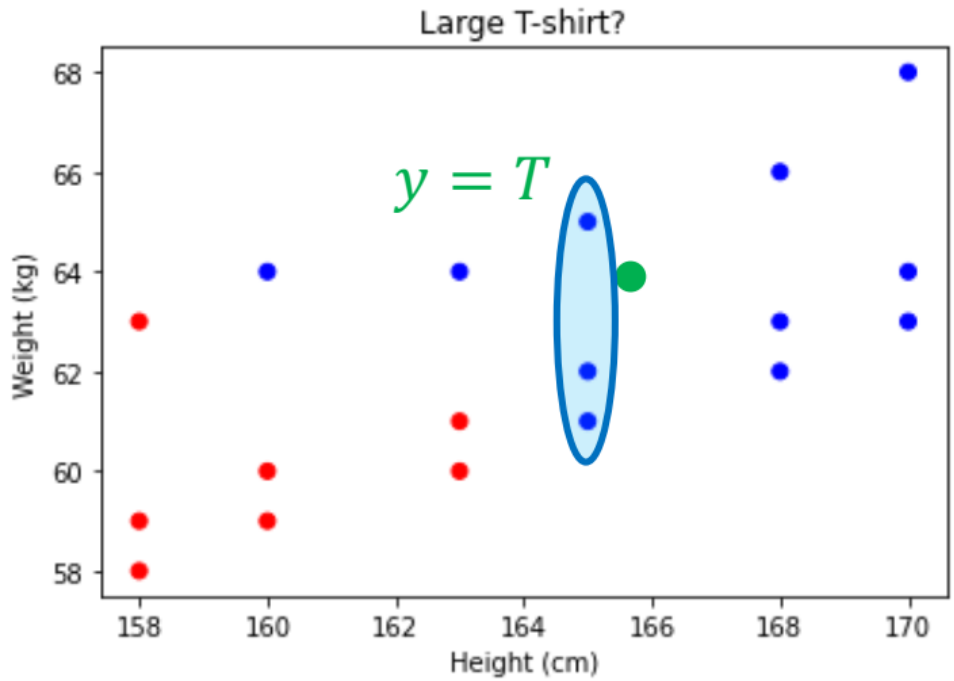
$$cari\ k\ buah\ tetangga\ terdekat\ \{i_1, \dots, i_k\}\ dari\ dataset$$
$$i_1, \dots, i_k = kArgMin_{i\ \in\{1,\dots,n\}} dist(x, x_i)$$

- Langkah-2:

$$Return\ label\ mayoritas$$
$$y = Majority\ \{y_{i1}, \dots, y_{ik}\}$$

Contoh

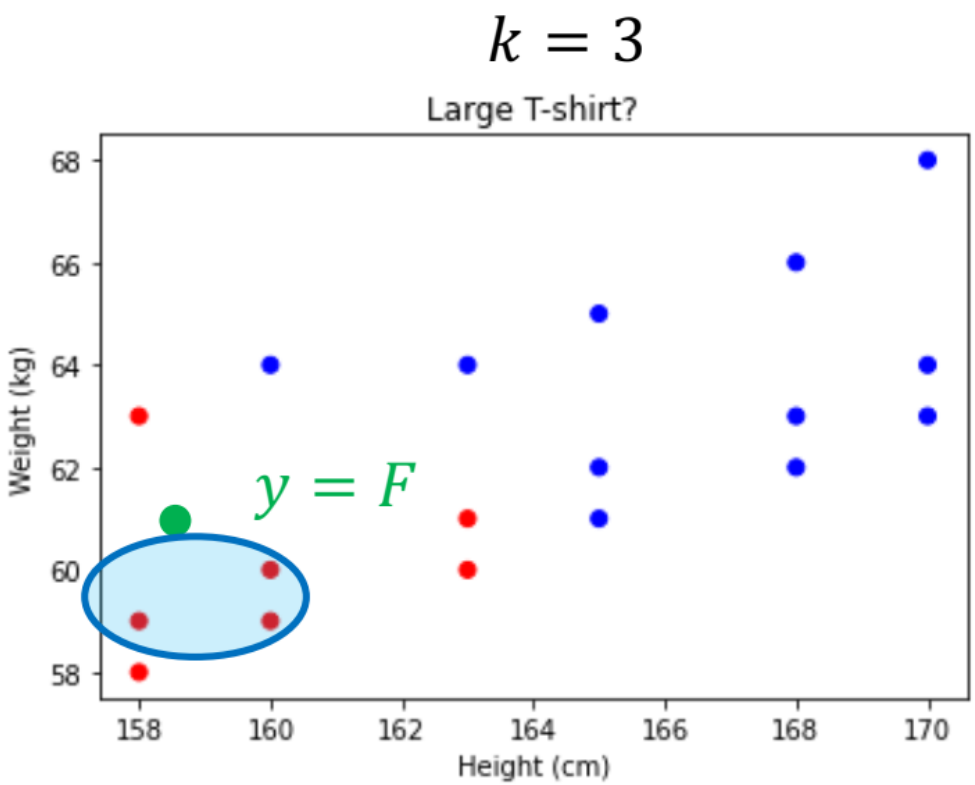
$k = 3$



Height (cm)	Weight (kg)	Large (vs Medium) t-shirt?
158	58	F
158	59	F
158	63	F
160	59	F
160	60	F
163	60	F
163	61	F
160	64	T
163	64	T
165	61	T
165	62	T
165	65	T
168	62	T
168	63	T
168	66	T
170	63	T
170	64	T
170	68	T

Inputs x_i Labels y_i

Contoh



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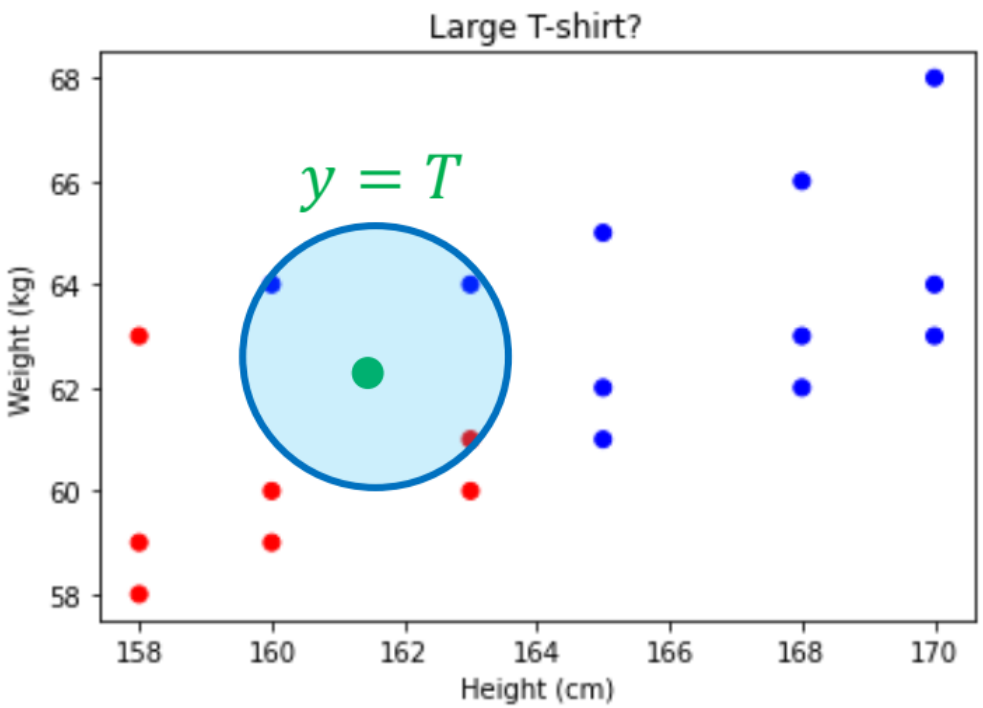

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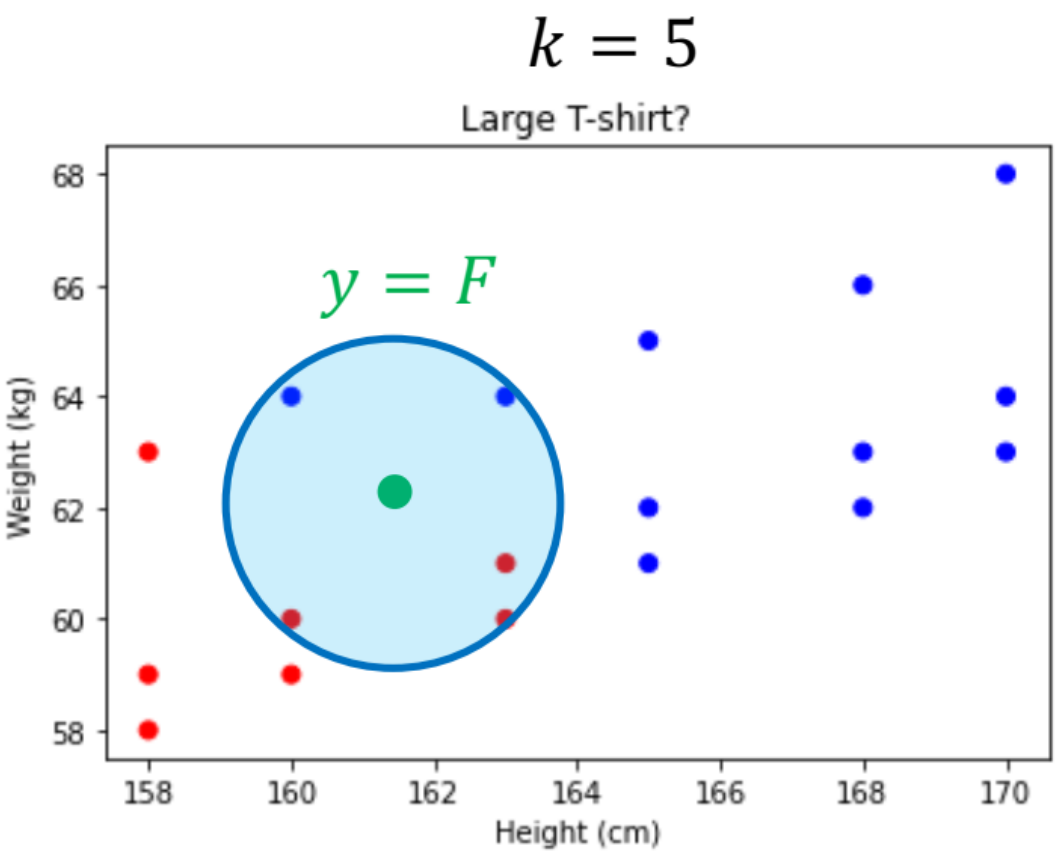

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Inputs x_i Labels y_i

kNN: Regression

- Regression: Diberikan input data x:
- Langkah-1:

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$$i_1, \dots, i_k = kArgMin_{i\ \in\{1,\dots,n\}} dist(x, x_i)$$

- Langkah-2:

$$Return\ the\ average\ label$$
$$y = Average\ \{y_{i1}, \dots, y_{ik}\}$$

- Pendekatan ini juga dapat digunakan utk menghitung probabilitas pada Classification

kNN: General Framework

- Design decision 1: Fungsi jarak apa yang digunakan?
- Design decision 2: Bagaimana melakukan agregasi label?

Fungsi Distance

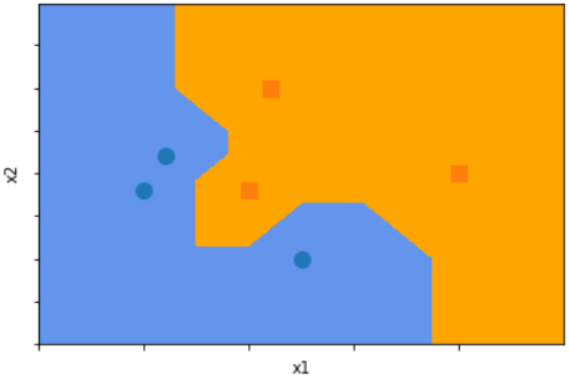
$\boldsymbol{L_1}$ **distance:** $\mathrm{dist}(x, x') = \|x - x'\|_1 = \sum_{j=1}^{d} |x_j - x_j'|$

$\boldsymbol{L_2}$ **distance:** $\mathrm{dist}(x, x') = \|x - x'\|_2 = \left(\sum_{j=1}^{d} (x_j - x_j')^2\right)^{\frac{1}{2}}$

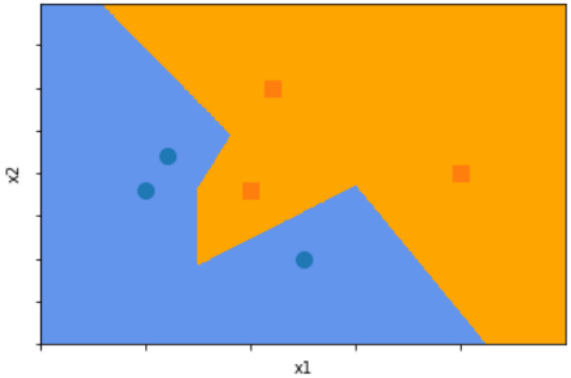
$\boldsymbol{L_\infty}$ **distance:** $\mathrm{dist}(x, x') = \|x - x'\|_\infty = \max_{j \in \{1,\ldots,d\}} |x_j - x_j'|$

$\boldsymbol{L_p}$ **distance:** $\mathrm{dist}(x, x') = \|x - x'\|_p = \left(\sum_{j=1}^{d} |x_j - x_j'|^p\right)^{\frac{1}{p}}$

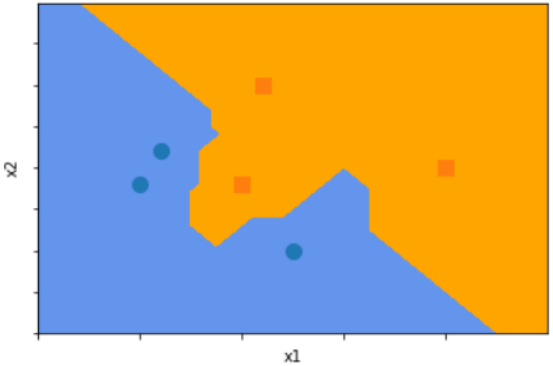
Fungsi Distance



L_1 distance



L_2 distance



L_∞ distance

Distance utk String

- Hamming distance: Jml karakter berbeda
  - Asumsi: Panjang karakter sama
  - Contoh: ABCDE vs AGDDF → Hamming Distance = 3
- Edit Distance: Jml operasi Insert/Delete/Replace yg dibutuhkan utk mengubah string menjadi string pembanding
  - Bisa menggunakan dynamic programming
  - Contoh: ROBOT vs BOT → Edit distance = 2

Karakteristik

- Kelebihan
  - Sangat sederhana
  - Nonparametric → Dapat melakukan learning terhadap decision boundaries yang kompleks
- Kekurangan:
  - Membutuhkan data yg exponential terhadap jml fitur
  - Waktu test/inferensi lama, karena proses perhitungan semuanya dilakukan saat itu,lazy learning, gak beneran belajar
- Pendekatan Modern:
  - Lakukan pada learned feature → Deep Learning
- kNN tidak melakukan minimisasi loss function secara langsung, namun bisa dievaluasi menggunakan standar loss (akurasi, MSE, dll)

Thank You